Marco Bellagente

Curriculum Vitae

Kaiserstr. 59, Heidelberg
✉ *marco.bellagente@gmail.com*
https://marcobellagente93.github.io

Education

2019- **Ph.D.**, *University of Heidelberg and Max Planck Institute for nuclear physics*.

2017-2018 **M.Sc. (Physics)**, *University of Milan*, 110/110 cum Laude.

2017-2018 **M.Sc.**, *Ludwig Maximilian Universität*, Exchange student.

Workshops and seminars

Conference **ICML 2021**, on-line, 2021 accepted for poster session.

School **EEML School 2021**, on-line, 2021 accepted for poster session.

Seminar **Turin particle physics group**, *Turin University*, on-line, 2021 invited speaker.

Workshop **ML4Jets**, *NY University*, NYC, 2020 accepted for talk.

School **A crash course in Machine Learning**, *IWR*, Heidelberg, 2019.

School **Nordic Probabilistic AI school**, *NTNU*, Trondheim, 2019.

Professional Skills

Coding Python, Pytorch, TF, LaTeX, shell-scripting, software versioning (Git-hub), C++, Mathematica, Jax

Languages English (fluent), German (intermediate), Italian (mother tongue)

Miscellaneous Monte Carlo simulations, numerical integration, analytical reasoning, strong mathematical background

Research and working interests

Generative models: normalizing flows/invertible neural networks and generative adversarial networks;
Refinement methods for generative models;
Topology methods for deep-learning;
Bayesian statistics and computation, Bayesian neural networks;
Neural networks as differential equations solvers;

Publications

2021 **Latent Space Refinement for Deep Generative Models**
R. Winterhalder, M. Bellagente and B. Nachman
`https://arxiv.org/pdf/2106.00792.pdf`, (submitted to NIPS)

2021 **Understanding Event-Generation Networks via Uncertainties**
M. Bellagente, M. Luchmann, M. Haußmann and T. Plehn,
`https://arxiv.org/pdf/2104.04543.pdf`, (accepted at INN+ workshop ICML 2021)

2020 **Invertible Networks or Partons to Detector and Back Again**
M. Bellagente, A. Butter, G. Kasieczka, T. Plehn, A. Rousselot, R. Winterhalder, L. Ardizzone and U. Köthe
`https://www.scipost.org/SciPostPhys.9.5.074`

2019 **How to GAN Away Detector Effects**
M. Bellagente, A. Butter, G. Kasieczka, T. Plehn and R. Winterhalder,
`https://www.scipost.org/SciPostPhys.8.4.070`

Personal interests

Hiking; jogging; gardening; reading; cooking (I often try classics from the Japanese cuisine. Results are debatable); studying German, complaining about my German.